TrES-2b

R-0022

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_260506` · created 2026-07-08T10:07:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461166.8835 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.043
Transit depth (Rp/Rs)^2	2.08 +/- 1.24 [%]
Orbital Inclination (inc)	88.57 +/- 2.89
Airmass coefficient 1 (a1)	1.69 +/- 0.93
Airmass coefficient 2 (a2)	-0.23 +/- 0.91
Scatter in the residuals of the lightcurve fit is	54.25 %
Best Comparison Star	#1 - [242, 159]
Optimal Aperture	0.87
Optimal Annulus	3.75
Transit Duration (day)	0.105 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.043 vs 0.1254 (deviation 0.43σ)
Duration fit vs published	0.105 d vs 0.0746 d (deviation 1.9σ)
Mid-time O–C	-2.7 min
Depth SNR	3.3
Residual scatter	54.25 %

Light curve

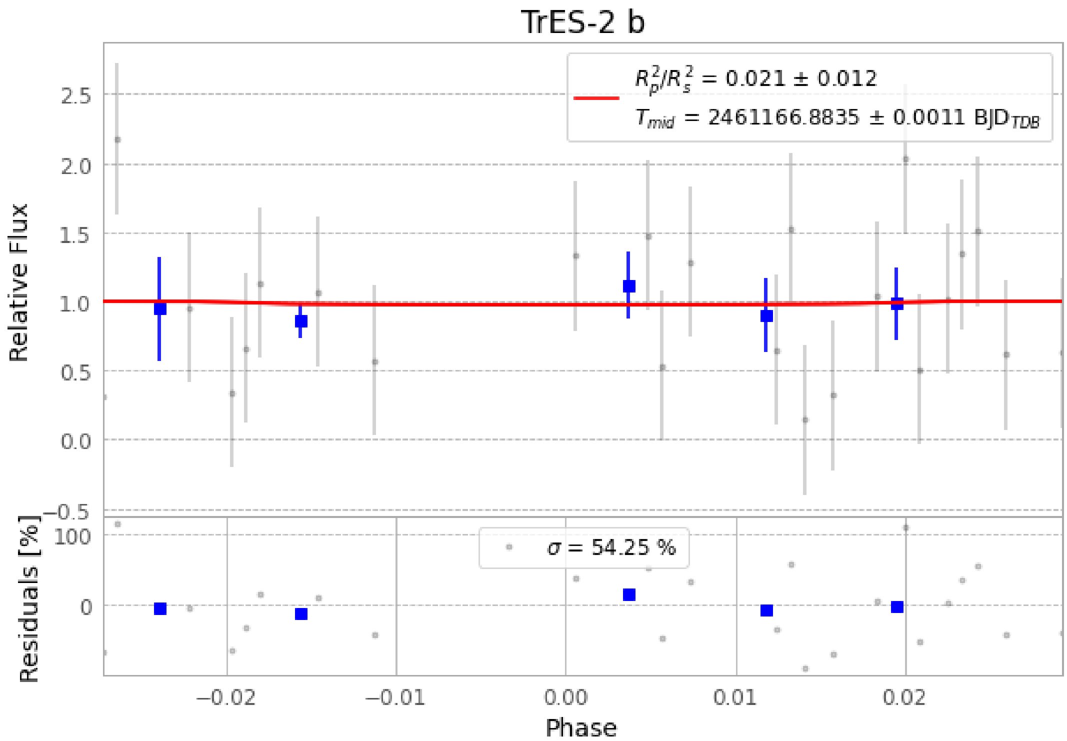

Figure 1 — FinalLightCurve_TrES-2 b_2026-05-06.png (SHA-256 4a992eaa2e860af7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 275], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3147d27747b8e6f4…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

75 FITS frames (49 MB), `TRES-2260506072415.FITS` → `TRES-2260506110615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-260506.log (e5e1b50db860…), FinalLightCurve_TrES-2 b_2026-05-06.png (4a992eaa2e86…), FinalLightCurve_TrES-2 b_2026-05-06.csv (a21259d7c450…)

Compute resources

Wall time 237.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)